

Lecture Schedule Integrated

PHYSICS – Track 01	
L. No.	Chapter and Sub topic
Vectors	
1.	Vectors- difference between scalars and vectors, graphical representation of vectors, unit vector, multiplication of scalars with vector (with graphical representation)
2.	vector addition (triangle law), Vector addition (parallelogram law), Analytical approach (both parallelogram and triangle law)
3.	Vector addition (polygon law), Problems on Vector addition, Vector Subtraction (analogous to vector addition), position vector and displacement vector as example
4.	Vector resolution (2D) for both any general vector and rectangular resolution, (3D resolution)
5.	Vector multiplication (dot product), angle between two vectors using dot product, resolution of vectors using dot product (2d & 3D)
6.	Vector multiplication (cross product)
7.	Module discussion
8.	Module discussion
Rectilinear Motion	
9.	General Kinematics: distance and displacement, speed and velocity (instantaneous and average), acceleration (instantaneous and average)
10.	Motion in 1 D: Uniform motion (basic examples), non uniform motion (constant acceleration),
11.	Graphical representation of motion,
12.	Motion in 1 D -Motion with constant acceleration, equations of motion.
13.	Problems on motion with constant acceleration
14.	Motion under gravity
15.	Motion with variable acceleration (application of calculus)
16.	Module discussion
17.	Module discussion
18.	Module discussion

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PHYSICS – Track 01	
L. No.	Chapter and Sub topic
Projectile Motion	
19.	Motion in 2D - Basic idea, introduction to types of projectile motion
20.	Motion in 2D - Oblique projectile motion (Trajectory, Hmax, Time of flight, Range)
21.	Module discussion
22.	Module discussion
23.	Motion in 2D - Inclined projectile motion (Projection up the incline)
24.	Motion in 2D - Inclined projectile motion (Projection down the incline)
25.	Module discussion
26.	Motion in 2D : Various non uniform acceleration cases
27.	Projectile Motion: Radius of curvature
28.	Module discussion
Relative Motion	
29.	Relative motion (1D various cases)
30.	Relative motion: River Boat problem
31.	Relative motion: Rain man, aircraft wind problem, Flag fluttering problems
32.	Relative Motion: Collision of projectiles
33.	Module discussion
Circular Kinematics	
34.	Circular Motion: Kinematics of circular motion, introduction to angular variables
35.	Basics of centripetal and tangential acceleration.
36.	Circular motion: relative angular velocity for particles
Work, Power and Energy	
37.	WPE: Calculation of work done by constant force
38.	Work done by variable force (integration and graphical approach), Work done by Spring
39.	Using path information to calculate work energy theorem

PHYSICS – Track 01	
L. No.	Chapter and Sub topic
Work, Power and Energy	
40.	Kinetic energy, work energy theorem and its application
41.	Module discussion
42.	Conservative and non-conservative forces (examples giving idea of perfect differential)
43.	Potential energy, Calculation of potential energy for continuous mass distribution
44.	Module discussion
45.	Conservation of mechanical energy, work energy theorem in terms of potential and kinetic energy
46.	Types of equilibrium
47.	Motion of particle along vertical circle
48.	Module discussion
49.	Module discussion
50.	Power
51.	Module discussion
52.	Module discussion
COM, Momentum and Collision	
53.	Center of mass (calculation for discrete particle system)
54.	Center of mass (calculation for continuous mass system)
55.	Center of mass with a part of the body been removed (concept of negative mass),
56.	Motion of cm,
57.	Conservation of linear momentum
58.	Thrust force (Variable mass systems)
59.	Impulse momentum theorem, impulse along a string
60.	Collisions (basics), coefficient of restitution, head on collision
61.	Module discussion
62.	Oblique collision
63.	Module discussion
64.	Module discussion
65.	Module discussion

PHYSICS – Track 01	
L. No.	Chapter and Sub topic
Rigid Body Mechanics	
66.	Rotational kinematics (concept of angular displacement, angular velocity, angular acceleration for a rigid body)
67.	Simultaneous translation and rotation for rigid body, calculation of linear variables from angular variables
68.	Kinematics of rolling
69.	ICR & IAR
70.	Moment of inertia (for discrete particle system), Moment of inertia for rigid bodies
71.	Moment of inertia for combination of rigid bodies, Theorem of parallel and perpendicular axes.
72.	Moment of inertia for rigid bodies with a part of mass being removed
73.	Module discussion
74.	Angular momentum (for discrete particle system and rigid bodies)
75.	Torque (about a point & about axis of rotation), newton's laws for rotation
76.	Application of newtons law's for various cases (equilibrium)
77.	Application of newtons law's for various cases (Pure rotation)
78.	Application of newtons law's for various cases (rolling)
79.	Module discussion
80.	Module discussion
81.	Work done due to a torque, kinetic energy due to simultaneous translation and rotation
82.	Work energy theorem and conservation of energy (based on problems for rotation, simultaneous translation and rotation)
83.	Module discussion
84.	Rotational impulse (including impulse due to friction)

PHYSICS – Track 01	
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Rigid Body Mechanics	
85.	Rigid body collisions (Using Conservation of angular momentum)
86.	Rigid body collisions (Using law of restitution)
87.	Module discussion
88.	Module discussion
Simple Harmonic Motion	
89.	SHM (kinematics of SHM), Phasor representation
90.	Module discussion
91.	SHM (determination of time period - force method)
92.	SHM (determination of time period - torque method)
93.	SHM (determination of time period - energy method)
94.	Superposition in SHM
95.	Damped and forced oscillation
96.	Module discussion
97.	Module discussion
Gravitation	
98.	Gravitation (for PU): Newton's law of gravitation, principle of superposition, Gravitation field (Acceleration due to gravity)
99.	Gravitation (for PU): acceleration due to gravity and its variation, gravitational potential energy (interaction), gravitational potential
100.	Gravitation (for PU): orbital speed and escape speed, Kepler's law
101.	Module discussion

PHYSICS – Track 02	
L. No.	Chapter and Sub topic
Basic Mathematics	
1.	Basic Mathematics - Solution to simultaneous equation (upto 3 variables), basic trigonometric formulas, inverse functions
2.	Binomial theorem and approximation, exponential and log functions and approximation, trigonometric approximation, coordinate geometry (introduction to slope, tangent and secant)
3.	Basic Calculus - introduction to limit (using coordinate geometry), differentiation (analysis of differentiation using limit and graph), differentiation (addition, subtraction, product rule, division rule)
4.	Differentiation (chain rule), Application of differentiation (rate of change and slope)
5.	Application of differentiation (maxima and minima)
6.	Integration (basic), integration by substitution
7.	Module discussion
8.	Definite Integral
9.	application (calculation of circumference, area and volume)
Units and Measurements, Errors and Experiments	
10.	Introduction - fundamental and derived physical quantities, units, characteristics of a unit, measurement of length, mass and time. rules to write the units. practical units. fundamental and derived units, mks, cgs and si units.
11.	Dimension and dimensional formulae, principle of homogeneity of dimensions, uses and limitations of dimensional analysis. Arbitrary choice of fundamental quantities and units. problems
12.	Accuracy, precession of instruments. significant figures-rules, concept and types of errors,
13.	Determination of relative and percentage errors of addition, subtraction, multiplication and division of quantities. problems

PHYSICS – Track 02	
L. No.	Chapter and Sub topic
Units and Measurements, Errors and Experiments	
14.	Least count-discussion on vernier callipers and screw gauge problems
15.	Module discussion
16.	Module discussion
Newton's Laws of Motion	
17.	Newton's laws of motion: I law, II law and III law - Examples
18.	Common forces in Mechanics, (Weight, normal force, tension), FBD - Examples
19.	General Method of problem solving using Newton's laws. Application for translational equilibrium cases
20.	NLM applied to accelerated bodies. Blocks in contact, Blocks connected by string,
21.	Frames of Reference, Concept of Pseudo force (Inertial force)
22.	Motion on inclined plane, motion of blocks connected by strings on inclined plane.
23.	Concept of Spring force
24.	Application of Newtons laws for non-equilibrium cases when bodies connected with string have different acceleration, length constraint.
25.	Shortcuts for writing String constraints (velocity/acceleration along string; T.x method, velocity of pulley as average of velocity of string on either side)
26.	Wedge constraints equation for movable multi-particle contact systems (geometrical method and relative motion along common normal method). General constraints.
27.	Module discussion
28.	Module discussion
29.	Module discussion
30.	Friction: Basic theory of friction, Static friction, limiting friction, kinetic friction and basic idea about rolling friction

PHYSICS – Track 02	
L. No.	Chapter and Sub topic
Newton's Laws of Motion	
31.	Angle of friction and repose, Block over block problems (using ground F.O.R and pseudo force)
32.	Module discussion
33.	Dynamics of circular motion and role of friction in circular motion, circular motion in horizontal plane, radius of curvature a path.
34.	Centrifugal force, Coriolis force, Banking of tracks
35.	Module discussion
Mechanical properties of Solids (Elasticity)	
36.	Definition of Elasticity, restoring forces, stress, strain, types of stresses and strains, Problems
37.	Hooke's Law, Young's Modulus, Bulk Modulus, Modulus of Rigidity, Elastic behaviour of materials. Stress - Strain graph
38.	Poisson's ratio. Elastic behaviour of materials, Experimental Determination of Young's Modulus (Searles Method)-Problems
39.	Stress in composite rods, Problems
40.	Stress in a composite wire, Analogy of Rod as a spring, Elastic Potential Energy
41.	Module discussion
Thermometry	
42.	Thermometry: Temperature and Heat, Zeroth law of thermodynamics, Measurement of temperature and various scales, Ideal gas equation and Absolute temperature, Thermometric property,
43.	CVGT, Different types of thermometers, Modern thermometry, Problems

PHYSICS – Track 02	
L. No.	Chapter and Sub topic
Thermal Expansion	
44.	Thermal expansion of solids – Coefficients of linear, superficial and volume expansion of solids (a, b, g) – Relation between a, b and g-Applications: (a) Expansion of cavities (b) Correction for length measured by a metallic scale
45.	Thermal expansion of solids – Error in time measured by a pendulum clock, Bending of bimetallic strips and Thermal stress
46.	Thermal expansion of liquids - Apparent and Real expansion of liquids – The coefficients of apparent expansion and real expansion of a liquid and the relation between them. Applications to (a) variation of density (b) variation of upthrust on a solid immersed in a liquid (e) Anomalous expansion of water – Problems
47.	Module discussion
Calorimetry	
48.	Calorimetry-Measurement of thermal energy - units for thermal energy - Thermal capacity - Specific heat - Determination of specific heat of a solid by the method of mixtures.
49.	Water equivalent, Phase transformation, Effect of pressure on melting point and boiling point, Regelation. Latent heat.
Heat Transfer	
50.	Heat Transfer: Conduction, Steady state heat flow, Heat current, Thermal Resistance,
51.	Radial and Cylindrical flow of heat.
52.	Conductors in series and parallel, Formation of Ice on lake
53.	Convection, Forced and natural convection, Sea breeze and land Breeze
54.	Thermal radiation, Position in Electromagnetic spectrum, Prevost's theory of exchanges, Emissive power and absorptive power of a body, Kirchhoff's law.

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Heat Transfer	
55.	Black body, Emissivity of a body, Blackbody radiation, Wein's law.
56.	Stefan-Boltzmann law, Newton's law of cooling, Newtons law from Stefan's law. Average form of Newton's Laws of Cooling.
57.	Module discussion
Kinetic Theory of Gases	
58.	Gas Laws, Kinetic theory of an ideal gas, Assumptions
59.	Maxwell Boltzmann distribution, rms, most probable and average speed
60.	Kinetic energy of gas molecule, microscopic interpretation of temperature
61.	Degrees of Freedom and Equipartition of Energy, Internal Energy of a gas, Equivalent degrees of freedom of gaseous mixture, mixing of gases at constant volume
62.	Mean free path, Mean collision time and Collision frequency
Thermodynamics	
63.	State of a gas and indicator diagram: Properties of indicator diagram, state variable and Path variable, cyclic process (positive and negative cycles), Heating a gaseous system and work done by a gas, change in internal energy of a gas on heating
64.	Thermodynamics: First law of thermodynamics, specific heat capacities of gases (molar and mass),
65.	Different type of Thermodynamic process: Isochoric, Isobaric, Isothermal, Adiabatic, free expansion (calculation of work done)
66.	Polytropic process: Standard process as a special case of Polytropic process, Indicator Diagram for Polytropic Process
67.	Molar specific heat of standard and some other processes

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L. No.	Chapter and Sub topic
Thermodynamics	
68.	Carnot's heat engine, and efficiency of other cyclic process
69.	Module discussion
70.	Refrigerators, Entropy and Second Law of thermodynamics: Reversible Process
71.	Module discussion
Fluid Mechanics	
72.	Hydrostatics: Density relative density (sp. Gravity), Density of mixture of different fluids, Pressure, variation of pressure in stationary liquid, Pascal's law and its application
73.	Variation of pressure due to accelerating container (horizontal and vertical acceleration)
74.	Barometer, variation in atmospheric pressure due to height
75.	Variation in density of liquid due to depth, Variation of pressure due to accelerating container (rotating container), equi-pressure lines
76.	Hydrostatic force on a flat and curved surfaces (vertical and horizontal) Torque due to hydrostatic force
77.	Buoyant Force, Point of action of hydrostatic (Buoyant) force (for torque calculation), Variation in buoyant force due to acceleration of container
78.	Module discussion
79.	Fluid dynamics: streamline and turbulent flow, REYNOLD'S NUMBER, Equation of continuity, Bernoulli's theorem
80.	Applications of Bernoulli's theorem- Torricelli's theorem and velocity of efflux
81.	Applications of Bernoulli's theorem, venturi meter, pitot tube, magnus effect etc
82.	Module discussion

PHYSICS – Track 02	
L. No.	Chapter and Sub topic
Fluid Mechanics	
83.	Viscosity: Effects of temperature on viscosity. Newton's law of viscosity.
84.	Stoke's law and terminal velocity
85.	Definition of surface tension. Molecular theory of surface tension. Surface energy and relation between surface tension and surface energy.
86.	Angle of contact, pressure difference across a liquid surface. Excess pressure inside a drop and a bubble.
87.	Capillary action, determination of surface tension by capillary rise
88.	Module discussion
Transverse waves (String Waves)	
89.	Basic definition, types of waves, Basic terminology for wave pulse and wave train (amplitude, frequency, displacement, wave velocity, particle velocity), Wave equation
90.	Simple Harmonic Wave equation for transverse waves, displacement, velocity and acceleration of particle from wave equation, relation between wavelength and frequency, Speed of Transverse wave along a stretched string and Power transmitted along a stretched string
91.	Module discussion
92.	Principle of Superposition, Reflection and Transmission of waves and free and fixed interface
93.	Standing Waves as application of principle of superposition, Normal modes of vibration of a stretched string, Sonometer and Melde's Experiment, Composite wires
94.	Module discussion

PHYSICS – Track 02	
L. No.	Chapter and Sub topic
Longitudinal Waves (Sound Waves)	
95.	Longitudinal waves, Expression of simple harmonic longitudinal waves in terms of displacement, Relation between displacement amplitude and pressure amplitude, Expression of simple harmonic longitudinal waves in terms of pressure and density, Speed of longitudinal waves in solids, liquids and gases
96.	Module discussion
97.	Sound waves, Newton's formula for sound waves, Laplace correction, factors effecting speed of sound, Power (and energy) propagation through longitudinal waves
98.	Intensity of wave, Loudness of sound waves, Application of principle of superposition: Interference of sound waves, Seebecks tube and Quincke's tube, Some other example of interference
99.	Module discussion
100.	Standing waves, Open and Closed organ pipes, Standing waves in clamped rod, Resonance tube and Kundt's tube, Beats, Doppler's effect in Sound
101.	Module discussion

CHEMISTRY - Track 01	
L.No.	Chapter / Sub topics
Atomic Structure	
1.	Introduction; Discovery of subatomic particles [Cathode + Anode Rays]
2.	J J Thomson and Rutherford's Model; Estimation of closest approach
3.	Some terms(Isotopes, Isobars, Isotones, Isolelectronic), Electromagnetic Radiations (properties of waves)
4.	Planck's Quantum Theory, Black Body Radiation
5.	Photoelectric Effect

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CHEMISTRY - Track 01	
L.No.	Chapter / Sub topics
Atomic Structure	
6.	Bohr's Atomic Model - Postulates, Applications of Bohr's model - Derivation of Expressions
7.	Explanation of Hydrogen Spectrum Using Bohr's model
8.	Problems on Bohr's Model and Applications
9.	De Broglie's Dual Wave Nature, Heisenberg Uncertainty Principle
10.	Schrodinger Wave Equation, Quantum Numbers
11.	Quantum Numbers Continued..,
12.	Concept of Nodes and Graphical representation
13.	Shapes of Orbitals
14.	Electronic configuration
15.	Module Problem solving session
Chemical Bonding	
16.	Types of bonding (Definitions of Ionic bond Covalent bond and Metallic bond) and octet rule, Limitations of octet rule, Formal charge
17.	Writing the Lewis dot structure
18.	Writing resonating structures, finding average bond order
19.	Stability of resonating structures, Finding bond order in oxo anions and their acids
20.	VBT, overlapping of orbital, Hybridisation
21.	VSEPR
22.	VSEPR
23.	Hybridization
24.	Hybridization
25.	Bond angle & Bond length / Bond Strength
26.	Type of p bonding (pp-pp & pp dp bond) & Coordinate bonding.

CHEMISTRY - Track 01	
L.No.	Chapter / Sub topics
Chemical Bonding	
27.	Electron deficient bonding & Back bonding
28.	Hydrogen Bonding
29.	Hydrogen Bonding
30.	Molecular Orbital Theory
31.	Application of Molecular Orbital Theory
32.	Application of Molecular Orbital Theory
33.	Metallic Bonding, Van der Waal's Forces & Fajan Rule
34.	Dipole moment
35.	Acidic and Basic Character
Gaseous States	
36.	Boyle's law, Charle's law, Gay-lussac's law, Avogadro's hypothesis
37.	Barometer & faulty barometer
38.	Ideal gas Equation, Connecting vessels problems
39.	Dalton's law and its applications.
40.	Graham's law of diffusion & effusion
41.	KTG
42.	Maxwell's distribution of gas velocities, Eudiometry
43.	Real gas Introduction, Vanderwaal's Equation & Verification
44.	Virial equation, Critical phenomena
45.	Reduced equation of state and Discussion
Liquids	
46.	Vapour Pressure, Boiling Point, Surface Tension, Viscosity
Thermodynamics	
47.	Definitions, Graph Conversion
48.	Reversible & Irreversible Process, Introduction of First Law
49.	Heat & Internal Energy
50.	Calculation of Work - Isothermal, Isochoric & Isobaric

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CHEMISTRY - Track 01	
L.No.	Chapter / Sub topics
Thermodynamics	
51.	Problems on FLOT, Work done
52.	Cp & Cv, γ , Enthalpy
53.	Adiabatic Process
54.	Questions based on Work + Comparison between Isothermal and Adiabatic
55.	Some other processes (Phase Transformation, Polytropic Process, Free Expansion, For solids & Liquids)
56.	Second Law (Basics of Entropy and Statement of 2nd law)
57.	Second Law (Entropy Calculation for various processes, Third Law of T.D.
58.	Problems based on Second Law of Thermodynamics
59.	Gibb's Free Energy and Gibb's Helmholtz Equation; Criteria of spontaneity
60.	Relation between Standard Gibb's Free Energy and Equilibrium constant;
Thermochemistry	
61.	Thermochemistry
62.	Thermochemistry
63.	Thermochemistry
Chemical Equilibrium	
64.	Introduction & Law of Mass Action; Characteristics of Equilibrium
65.	Equilibrium Constants (K_p & K_c), Questions based on K_p & K_c
66.	Reaction Quotient & Characteristics of equilibrium constant.
67.	Degree of Dissociation and Heterogenous Equilibrium
68.	Vapour Pressure, Relative Humidity, Thermodynamics of Equilibrium
69.	Le-Chatelier principle: Effect of concentration, pressure, temperature, adding inert gases

CHEMISTRY - Track 01	
L.No.	Chapter / Sub topics
Chemical Equilibrium	
70.	Le-Chatelier principle: Physical Equilibria, Problems
71.	Simultaneous Equilibrium
Ionic Equilibrium	
72.	Concept of acids and bases : Definitions, drawbacks
73.	Properties of water; Levelling Solvent
74.	pH calculation (SA, SB, mixture of SA, mixture of SB, mixture of SA & SB)
75.	pH Calculation of WA, WB (Ostwald's dilution law) ;
76.	pH calculation of Mixture of Weak acids, Weak bases, Strong acid + Weak acid, Strong base + Weak base
77.	pH calculation of polyprotic WA/WB and their mixtures
78.	Salt hydrolysis (WA + SB, SA + WB, WA + WB)
79.	Salt hydrolysis (Salts of polyvalent ion and amphiprotic ion)
80.	Buffer Solutions, pH calculation of buffer solutions (generated from Monobasic acid/monoacidic base)
81.	Buffer Solutions (pH calculation of Polyacidic base & polybasic acid) + Indicators
82.	Solubility (K _{sp} + Effect of common ion on solubility + Condition for Precipitation)
83.	Solubility (Selective precipitation, Solubility exchange)
84.	Problems on Solubility Product
85.	Titration (Monoacidic bas)
86.	Titration (Polybasic acid)
87.	Titration (Polyacidic base & polybasic acid)
88.	Titration (Polyacidic base & polybasic acid)
Redox Reactions	
89.	Oxidation Number
90.	Balancing redox reactions

CHEMISTRY - Track 01	
L.No.	Chapter / Sub topics
Redox Reactions	
91.	Balancing redox reactions
92.	Redox Reactions - Concept of Electrode Potential (NCERT Aspects); Galvanic Cell
Equivalent Concept and Volumetric Analysis	
93.	Introduction to Equivalent concept - definition, methods of determining Equivalent weight
94.	Normality, Equivalent weight calculations in acids, bases, salts and redox reactions; Drawbacks of Equivalent concept
95.	Volumetric analysis - Terminologies, Principle, Types of titration
96.	Acid Base Titration
97.	Double Indicator, Problems based on Acid Base Titration
98.	Redox Titration - Permanganometric, Dichrometric Titration
99.	Iodometric and Iodimetric Titrations
100.	Iodometric and Iodimetric Titrations
101.	buffer session

CHEMISTRY - Track 02	
L.No.	Chapters/Sub topics
Some Basic Concepts of Chemistry (Mole Concept)	
1.	Introduction; Measurement (Basic and SI units), uncertainty in measurement
2.	Laws of Chemical Combination
3.	Scientific notation, Significant figures
4.	Basic definitions : amu , GMM , GAM, mole , Avogadro's number
5.	Mole- mass number conversion for atoms or molecules. $PV = nRT$ & Question based on it, STP.
6.	Density, % Composition by mass, by mole, Minimum Molecular Mass Determination

CHEMISTRY - Track 02	
L.No.	Chapters/Sub topics
Some Basic Concepts of Chemistry (Mole Concept)	
7.	Empirical & Molecular Formula, Stoichiometry
8.	Equation based calculations, Concept of Limiting reagent
9.	% Excess, % Yield, POAC
10.	Sequence & Parallel Reactions, Mixture analysis & % Purity
11.	Concentration terms & their interconversion
12.	Dilution and Mixing
13.	Problem Solving Session
Periodic Classification	
14.	General Introduction of Periodic Table, Effective Nuclear Charge (Z effective)
15.	Atomic & Ionic Radius & Lanthanoid contraction
16.	Ionization Energies & Electron Gain Enthalpies & Electro negativity
17.	Electron Gain Enthalpies & Electro negativity
18.	Periodicity in Chemical Properties
IUPAC Nomenclature	
19.	Introduction to Covalent Bond, Introduction of s and p bond. Valencies of C, H, X, O & N, concept of hybridization, Basic definitions - Electronegativity
20.	Degree of C, H & X and structure formula, Hybridization of carbon (elementary idea) & DU of Hydrocarbon only, General formula, Bond Line formula, Structural formula and homologs
21.	Classification of organic compound, aromatic compound (excluding Huckell rule) bicyclic compound, spiro compound. General Rules of IUPAC Nomenclature
22.	IUPAC-Nomenclature of Alkane & Cycloalkane with simple side chain (Alkyl Radical)
23.	IUPAC-Nomenclature of Alkane & Cycloalkane with complex alkyl radical

CHEMISTRY - Track 02	
L.No.	Chapters/Sub topics
IUPAC Nomenclature	
24.	IUPAC-Nomenclature of Alkenes, alkynes, Cycloalkene and polyene
25.	IUPAC-Nomenclature of non-chain terminating Functional groups (–OH, –SO ₃ H, Ketone, thiols and amines)
26.	IUPAC-Nomenclature of chain terminating Functional groups (Aldehyde & Carboxylic acids, (Amides, Oyl halide & Nitriles)
27.	IUPAC-Nomenclature of chain terminating Functional groups (Esters & Anhydride)
28.	IUPAC-Nomenclature of aromatic compounds & Discussion of IUPAC
29.	Structural isomerism, DU calculation of Hetero atom (N,O,S & X) containing compounds
30.	Isomer counting of Hydrocarbon, Isomer counting of Hetero atom (N,O,S & X) containing compounds
31.	Problem Solving Session
Physical properties, Qualitative analysis and Quantitative analysis	
32.	Purification of organic compounds
33.	Qualitative analysis and Quantitative analysis
34.	Qualitative analysis and Quantitative analysis
35.	Qualitative analysis and Quantitative analysis
GOC- I	
36.	Inductive effect
37.	Resonance effect (Drawing Structure)
38.	Stability of Resonating structure
39.	Mesomeric effect & SIR
40.	Hyperconjugation
41.	Aromaticity
42.	Application of I.R.,M HC effects (Bond Length & Electron density on benzene ring)
43.	Factors affecting stability of carbocations

CHEMISTRY - Track 02	
L.No.	Chapters/Sub topics
GOC- I	
44.	Rearrangement of Carbocations
45.	Factors affecting stability of carbanions
46.	Free radicals - Factors affecting stability of free radicals;
GOC- II	
47.	Basic Strength
48.	Basic Strength
49.	Acidic Strength
50.	Acidic Strength
51.	Tautomerism
52.	Problem Solving Session
Stereochemistry	
53.	Stereochemistry – Projection Formulae (Wedge Dash, Fischer Projection, Newmann and Saw Horse Projection
54.	Interconversion of Projection Formulae
55.	Configurational isomerism - geometrical isomerism ;
56.	Conformational analysis - ethane and butane; Bayer Strain Theory,
57.	Conformational analysis - Cyclohexane;
58.	Optical Isomerism - Conditions for showing optical activity
59.	Enantiomers, diastereomers, racemic mixture, problems
60.	Optical isomerism due to one or more chiral centers (carbon/nitrogen); Optical isomerism due to chiral axis
Hydrocarbons	
61.	Alkanes - Introduction, Isomerism
62.	Alkanes - Preparation of alkanes
63.	Alkanes - Preparation of alkanes continued...
64.	Reactions of alkanes - Mechanism of free radical substitution (halogenation of alkanes)

CHEMISTRY - Track 02	
L.No.	Chapters/Sub topics
Hydrocarbons	
65.	Reactions of alkanes - Aromatization, Isomerization, Oxidation, Combustion and Pyrolysis
66.	Alkenes - Introduction, Isomerism
67.	Alkenes - Preparation
68.	Alkenes - Preparation Continued..,
69.	Reactions of alkenes - Halogenation of alkenes (with mechanism along with stereochemistry), addition of Bromine water, HOCl
70.	Reactions of alkenes - Hydrohalogenation, Peroxide effect, Hydrogenation (syn addition)
71.	Reactions of alkenes - Hydration, Oxymercuration - demercuration, Hydroboration - oxidation
72.	Reactions of alkenes - Ozonolysis, Oxidation of alkenes, Hydroxylation - syn and anti
73.	Alkynes - isomerism, preparation
74.	Alkynes - Electrophilic addition reactions
75.	Alkynes - Reactions due to terminal alkyne; Test to distinguish between alkanes and alkenes / alkynes
76.	Aromatic Hydrocarbons - Classification, Aromaticity, Preparation of Benzene
77.	Mechanism of electrophilic substitution reactions - halogenation and nitration
78.	Friedel - Craft's alkylation and acylation reactions - Mechanism and Problems
79.	Mechanism of sulphonation;
80.	Orienting influence towards electrophilic substitution reactions; Problems on Electrophilic substitution
Hydrogen	
81.	Position of hydrogen in periodic table, occurrence, isotopes, preparation. Properties and uses of hydrogen, hydrogen economy.

CHEMISTRY - Track 02	
L.No.	Chapters/Sub topics
Hydrogen	
82.	Hydrides-ionic, covalent and interstitial with one example for each. Physical properties of water.
83.	Chemical properties of water (amphoteric nature, reaction with Na metal), heavy water, Hardness of water-temporary and permanent methods of removing hardness of water..
84.	Hydrogen peroxide – preparation from BaO_2 oxidising property – with PbS , in acidic medium, reducing property with I_2 , structure and uses, volume strength of H_2O_2 and problems.
s-Block Elements	
85.	Introduction, Group - 1: Atomic Properties and Physical Properties
86.	General Chemical Properties of Group - 1
87.	Compounds of Sodium: Sodium oxide, sodium peroxide, NaOH ; Sodium carbonate and other compounds of Group - 1; Biological importance of sodium and potassium ions
88.	Introduction, Group - 2: Atomic Properties and Physical Properties
89.	General Chemical Properties of Group - 2
90.	Compounds of Calcium; other compounds of Group - 2
Environmental Chemistry	
91.	Environmental pollution – air, water and soil pollution, chemical reactions in atmosphere, smog, major atmospheric pollutants.
92.	Effect of air pollution. Acid rain, ozone and its reactions, effects of depletion of ozone layer, greenhouse effect and global warming.
93.	Pollution due to industrial wastes, green chemistry as an alternative tool for reducing pollution, strategy for control of environmental pollution

CHEMISTRY - Track 02	
L.No.	Chapters/Sub topics
p-Block Elements	
94.	Introduction to Group 13: Atomic Properties, Physical Properties
95.	General Chemical Properties of Group - 13
96.	Compounds of Boron: Boron, Boron trioxide, Boric acid, Borax
97.	Compounds of Boron: Diborane; Halides of boron, Compounds of Al: Alumina, Aluminium chloride, Alum; Anomalous Properties of Boron;
98.	General Chemical Properties of Group - 14
99.	Carbon - Properties, Allotropic forms of carbon; Compounds of carbon
100.	Silicon - Properties, Compounds of Silicon; Halides and Oxides of Tin and lead;
101.	buffer session

MATHEMATICS - Track 01	
L.No.	Chapter / Sub topics
Basic Maths	
1.	Theory of numbers-natural, whole numbers, Integers, rational and irrational numbers, real numbers, prime numbers, LCM and HCF of numbers and fractions, Problems
2.	Ratio and Proportion, Fractions, continued fractions, concept of Invertendo, Alternendo, Componendo, Dividendo, Componendo and Dividendo, Problems. Rational algebraic fractions, Resolving rational algebraic fractions into its partial fractions, Problems.
3.	Concept of divisibility of algebraic expressions, Factor and remainder theorem, Solution of equation involving one and two unknowns, Problems.

MATHEMATICS - Track 01	
L.No.	Chapter / Sub topics
Basic Maths	
4.	Laws of indices, surds, rationalization, computing roots of a surd, Principle of mathematical induction, Problems
5.	Basic concepts in Geometry-Concepts pertaining to Similar, Congruent triangles. Perimeter & Area of triangle, Square, Rectangle, Parallelogram, Rhombus and Circle. Surface area and Volume of Cube, Cuboid, Sphere, Cylinder and Cone, Problems.
Trigonometric Ratios and Identities	
6.	Measurement of an angle, Basic results, Trigonometric ratios of an angle and identities, Problems
7.	Trigonometric ratios of standard angles, Circular trigonometric functions, Problems
8.	Problems continued
9.	Sign of trigonometric functions in quadrants, Trigonometric ratios of Allied angles, Graphs, Domain and Range of trigonometric functions, Problems
10.	Problems continued
11.	Trigonometric ratios for Compound Angles, Problems
12.	Problems continued
13.	Multiple and Sub multiple Angles, Problems
14.	Problems continued
15.	Values of Trigonometric ratios of Special angles - 18° , 36° , etc, Problems
16.	Problems continued
17.	Transformation Formulae, Problems
18.	Problems continued
19.	Sums of Sines, Cosines of n angles in A.P, Problems
20.	Conditional Identities, Problems
21.	Problems continued
22.	Heights and Distances, Problems
23.	Miscellaneous problems

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MATHEMATICS - Track 01	
L.No.	Chapter / Sub topics
Trigonometric Equations and Inequalities	
24.	Principal and General Solutions of Trigonometric equations, PU Problems
25.	General Solution of equations of type $\sin^2 x = \sin^2 \alpha$ etc. General Solution of equations of the type $a \cos x + b \sin x = c$, Problems
26.	Finding range of a parameter involved in given Trigonometric equation, Problems.
27.	Solving Trigonometric equations based on extreme values of given functions, Problems.
28.	Solving Trigonometric equations / inequations and transcendental equations by graphs, Problems
29.	Miscellaneous problems
30.	Buffer session
Properties of Triangles	
31.	Sine, Cosine, Projection rules, Law of tangents, Problems
32.	Problems continued
33.	Half angle formulae, Area of a Triangle, Problems
34.	Circumcentre, Incentre, Centroid, Orthocentre of a triangle, Relation between them, Problems
35.	Problems continued
36.	Solution of a triangle, Problems
37.	Problems
Introduction to Co-ordinate Geometry	
38.	Introduction, Distance formula, Section formula and its applications-Problems.
39.	Area of Triangle/Polygon, Problems
40.	Coordinates of different centers of a triangle, Problems.
41.	Problems
42.	Locus and its equation, Problems.

MATHEMATICS - Track 01	
L.No.	Chapter / Sub topics
Straight Lines	
43.	Slope of a line, Different forms of a line, Problems
44.	Problems continued
45.	Position of points relative to a line, position of a point w.r.t. two lines and related results, Problems
46.	Angle between lines, Condition for lines to be Perpendicular and parallel, Problems
47.	Distance of a point from a line, Distance between two parallel lines, Problems
48.	Problems continued
49.	Area of parallelogram, Problems
50.	Image of a point w.r.t a line, Problems
51.	Family of Lines, Concurrency of three lines, Problems
52.	Bisectors of angles between lines, related results and, Problems
53.	Problems
54.	Problems on Centres referred to a triangle
55.	Miscellaneous problems
Introduction to 3D Geometry PU	
56.	Introduction, Distance Formula, Collinearity of points, Section formula and Problems
Pair of Lines	
57.	Homogeneous equation of second degree in two unknowns representing a pair of lines through origin. Angle between them, Equation of angular bisectors between them, Problems
58.	Shifting of origin, Rotation of axis, Problems.
59.	General Equation of Second degree- Condition for it to represent a pair of lines, Point of intersection, Problems
60.	Problems continued
61.	Pair of parallel lines - condition, distance between them, Problems
62.	Concept of Homogenization, Problems
63.	Miscellaneous problems

MATHEMATICS - Track 01	
L.No.	Chapter / Sub topics
Circles	
64.	Definition of Circle - Standard form of circle - General Form of Circle - Diametric form of Circle, Problems
65.	Problems
66.	Intercepts made by circle, equation of the circle touching the axes, Parametric form of Circle, Problems
67.	Problems
68.	Position of point, with respect to the circle, Position of the line with respect to the circle, Length of the tangent, Problems
69.	Problems
70.	Equation of tangent in cartesian form, Parametric form and Slope form
71.	Equation of Director Circle, Equation of Tangent drawn from an external point and Angle between them
72.	Problems
73.	Common chord of two circles, Chord of Contact, Chord with given middle point
74.	Problems
75.	Position of Two Circles, Numbers of common tangents, Problems
76.	Radical Axis, Properties associated. Radical Centre, Angle between two circles, Orthogonality of Circles
77.	Problems
78.	Family of circles, Limiting points
79.	Problems
80.	Miscellaneous problems
Conic Sections PU	
81.	Introduction to Conic Sections, Focus- directrix property, General equation of degree two in x and y representing conics. Parabola- Derivation in Standard form, Parameters associated with it. Problems
82.	Problems on Parabola
83.	Ellipse & Hyperbola- Derivations in standard form, Parameters associated with them.
84.	Problems on Ellipse and Hyperbola

MATHEMATICS - Track 01	
L.No.	Chapter / Sub topics
Statistics	
85.	Measures of Central Tendency: Mean, Mode, Median, Properties, Problems
86.	Problems continued
87.	Measures of Dispersion: Range, Coefficient of range, Mean deviation, Quartile deviation, Properties, Problems
Statistics	
88.	Standard deviation and Variance, Coefficient of Variation, Properties, Problems
89.	Problems
Limits PU	
90.	Introduction to Limits, Algebra of limits, Limits of polynomial and Rational functions
91.	Limits of Trigonometric functions, Sandwich Theorem - Problems
92.	Problems continued
Methods of Differentiation	
93.	Definition of derivative of a function, Derivatives of functions from first principles, Algebra of derivatives of functions. PU Problems
94.	Derivatives of Composite functions, Chain Rule, Problems
95.	Problems Continued
96.	Differentiation of Implicit Functions, Parametric functions, Differentiation of one function w.r.t the other, Problems
97.	Problems
98.	Logarithmic differentiation, Higher order Derivatives, Problems
99.	Differentiation of Functional Relations
100.	Miscellaneous Problems
101.	Buffer session

MATHEMATICS - Track 02	
L.No.	Chapter /Sub-Topic
Sets	
1.	Set-Definition, Representation, Types, Subsets, Power Set, Universal Set, Venn Diagrams, Intervals, Problems.
2.	Operations on sets, Properties on operations of sets, Difference of sets, practical problems on Union and Intersection, Problems
3.	PU Problems
Relations	
4.	Cartesian Product, Definition of Relation, Types of Relations - PU problems
5.	Reflexive, symmetric and transitive relations, Equivalence relation, Problems
6.	PU Problems
Introduction to Functions (PU)	
7.	Definition, Types of functions, Some important functions and their graphs (Identity Function, Constant function, Polynomial function, Rational function, Modulus function, Signum function & Greatest integer function), Problems.
8.	Algebra of real functions, Domain & Range of a function, Problems.
Linear Inequalities PU	
9.	Inequalities Definition, Algebraic Solution of inequalities in one variable, Problems, Graphical Solution of linear inequalities in two variables, Problems
Fundamentals of Mathematics-1 & 2	
10.	Methods of solving rational algebraic inequalities -Wavy curve method, Problems
11.	Problems continued
12.	Problems continued
13.	Logarithmic form and exponential form of a number, Associated results, Problems.
14.	Problems continued
15.	Logarithmic and Exponential Functions- Graphs and Properties
16.	Problems on Logarithmic equalities and exponential equalities
17.	Logarithmic and Exponential Inequalities, Problems

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MATHEMATICS - Track 02	
L.No.	Chapter /Sub-Topic
Fundamentals of Mathematics-1 & 2	
18.	Problems
19.	Methods of solving irrational algebraic inequalities, Problems
20.	Problems continued
21.	Modulus Functions - Properties, Solution of equations involving Modulus functions
22.	Solution of Inequalities involving Modulus functions
23.	Miscellaneous Problems
24.	Buffer session
Quadratic Equations and Expressions	
25.	Quadratic Equations, Roots of a Quadratic Equation, Nature of roots of a Quadratic Equation, Problems
26.	Relation between roots and coefficients of a Quadratic equation, Problems
27.	Problems
28.	Symmetric Functions, Problems
29.	Quadratic expression and its Graphs -Sign of quadratic expressions, Maximum and Minimum value of a quadratic expressions, Problems
30.	Problems continued
31.	Location of Roots, Problems
32.	Problems continued
33.	Quadratic equations having a common root, common roots, Problems
34.	Relation between roots and coefficients of a Cubic equation, Biquadratic equation and an nth degree equation, Problems
35.	Problems
36.	Formation of a Polynomial equation, Descartes Rule of Signs, Position of roots of a Polynomial Equation, Problems
37.	Transformation of equations, Problems
38.	Miscellaneous Problems
39.	Buffer session

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MATHEMATICS - Track 02	
L.No.	Chapter /Sub-Topic
Sequence and Series	
40.	Arithmetic Progression, Properties, Problems
41.	Geometric Progression, Properties, Problems
42.	PU Problems
43.	Harmonic Progressions, Properties, Problems
44.	Insertion of AM, GM and HM, Properties on means, Problems
45.	Problems on inequalities involving AM, GM and HM
46.	Combined problems on AP, GP, HP
47.	Problem continued
48.	Arithmetic - Geometric sequences, Problems
49.	Summation of special series-Telescopic series etc, Problems
50.	Miscellaneous Problems on Summation of Series
51.	PU Problems
52.	Buffer session
Binomial Theorem for any index	
53.	Binomial theorem for positive integral index, Properties, General Term, Middle Term, Problems
54.	Greatest Term and Greatest Coefficients, Problems
55.	Problems continued
56.	Properties on Binomial Coefficients, Problems
57.	Integral and fractional parts of a Binomial expansion, Problems
58.	Multinomial Theorem, Properties, Problems
59.	Binomial Theorem for any index, Problems
60.	Problems
61.	Exponential and logarithmic series, Problems
62.	Problems
63.	Miscellaneous Problems
64.	Buffer session

MATHEMATICS - Track 02	
L.No.	Chapter /Sub-Topic
Permutations and Combinations	
65.	Permutations and Combinations- Fundamental Principle of counting, Linear Permutations of objects without repetition, Problems
66.	Linear permutation of objects with repetition, Permutation of n objects of which p are of one kind, q are of second kind and so on, Problems.
67.	Problems continued
68.	Circular Permutation, Related results, Problems
69.	Combinations of n distinct objects, Problems
70.	Combinatorial geometry, Problems
71.	Concept of generating functions, Combination of n objects of which p are of one kind, q are of second kind and so on, Problems.
72.	Combination of identical objects taken one or more objects at a time, Number of divisors of N , Sum of divisors of N , Problems
73.	Division and distribution of distinct objects into two or more groups, Problems
74.	Problems continued
75.	Division and distribution of identical objects into two or more groups, Problems
76.	Problems continued
77.	Principle of inclusion and exclusion, Number of functions, Number of one-one and onto function on two sets, Problems
78.	Problems continued
79.	Derangements, Problems
80.	Miscellaneous Problems
81.	Buffer session
Functions	
82.	Functions, Domain, Co-domain, Range of a real valued function
83.	One-one, onto, into and many to one functions, number of functions
84.	Classifications of functions, some special functions
85.	Domain, Rules for finding the domain, Problems
86.	Problems continued
87.	Range of a function, Rules for finding the range, Problems
88.	Problems continued
89.	Homogenous functions, Bounded functions, the range, Problems

MATHEMATICS - Track 02	
L.No.	Chapter /Sub-Topic
Functions	
90.	Periodic functions, Properties of periodic functions, Problems
91.	Composite functions, Properties, Problems
92.	Problems continued
93.	Inverse of a function, Problems
94.	Problems
95.	Functional equations, Problems
96.	Problems on functional equations
97.	Miscellaneous Problems
98.	Buffer session
Probability PU	
99.	Definitions of Random experiment, Sample space, event, simple & mixed events, Mutually exclusive events, Exhaustive events, Definition of probability, Addition theorem, PU Problems
100.	PU Problems
101.	Buffer session

Test Schedule

SL No	Test	Test Date
1	Weekly Test-Physics-1	27-Apr-26
2	Weekly Test-Chemistry-1	04-May-26
3	Weekly Test-Maths-1	11-May-26
4	Main Test-1	24-May-26
5	Weekly Test-Physics-2	01-Jun-26
6	Weekly Test-Chemistry-2	08-Jun-26
7	Main Test-2	21-Jun-26
8	Weekly Test-Maths-2	29-Jun-26
9	Weekly Test-Physics-3	06-Jul-26
10	Main Test-3	19-Jul-26
11	Advanced Test-1 (One paper)	19-Jul-26
12	Weekly Test-Chemistry-3	27-Jul-26
13	1st Quarterly Exam-1 & CET-1	3 Aug to 7th Aug 26
14	Main Test-4	23-Aug-26
15	Advanced Test-2 (One paper)	23-Aug-26
16	Weekly Test-Maths-3	31-Aug-26
17	Weekly Test-Physics-4	07-Sep-26
18	Main Test-5	19-Sep-26
19	Advanced Test-3 (Two papers)	20-Sep-26
20	Midterm Exam & CET-2	25 Sep to 9th Oct 26
21	Weekly Test-Chemistry-4	02-Nov-26
22	Main Test-6	21-Nov-26
23	Advanced Test-4 (Two papers)	22-Nov-26
24	Weekly Test-Maths-4	30-Nov-26
25	2nd Quarterly Exam & CET-3	10 Dec to 15 Dec 26
26	Weekly Test-Physics-5	21-Dec-26
27	Main Test-7	02-Jan-27
28	Advanced Test-5 (Two papers)	03-Jan-27
29	Weekly Test-Chemistry-5	11-Jan-27
30	Weekly Test-Maths-5	18-Jan-27
31	Main Test-8	27-Jan-27
32	Advanced Test-6 (Two papers)	28-Jan-27

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Holiday list- Bengaluru / Mysuru

Sl.No.	Date	Days	Holidays
1.	28 May 26	Thursday	Bakrid
2.	15 Aug 26	Saturday	Independence Day
3.	21 Aug 26	Friday	Varamahalakshmi Vrata
4.	14 Sep 26	Monday	Ganesha Chathurthi
5.	2 Oct 26	Friday	Gandhi Jayanthi
6.	10 Oct 26	Saturday	Mahalaya Amavasye
7.	19 Oct 26	Monday	Dasara Holidays may vary for Mysuru
8.	20 Oct 26	Tuesday	
9.	21 Oct 26	Wednesday	
10.	9 Nov 26	Monday	Deepavali Holidays
11.	10 Nov 26	Tuesday	
12.	25 Dec 26	Friday	Christmas